DAVID LÓPEZ DIEGO

Materials Characterization & Metrology Scientist | Nanomaterials & Scientific Data Analysis

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PROFESSIONAL SUMMARY

Physicist with 3+ years of experience in advanced materials characterization, nanoscale metrology, scientific instrumentation and experimental data analysis. Hands-on background across nanomaterials, semiconductors, biosensors, thermoelectric materials and biological systems, combining laboratory work with Python workflows for reproducible data processing and technical reporting. Experienced in multi-institutional R&D projects, instrument operation, calibration and quality documentation, with 10+ peer-reviewed publications. PhD candidate in Materials Physics, Nanoscience and Biophysics.

TECHNICAL SKILLS

Materials Characterization	AFM, SEM-EDX, TEM, XRD, Raman, FTIR, XPS, FIB lithography, optical & fluorescence microscopy, HPLC-MS, ICP
Data Analysis & Programming	Python, NumPy, Pandas, Scikit-learn, R, Excel, Origin, ImageJ, Git
Instrumentation & Lab Ops	Calibration, preventive maintenance, data acquisition, sample preparation, bacterial cultures, equipment logs
Quality & Documentation	GLP/GMP principles, ISO 9001, workflow standardization, procurement and budget tracking
Additional Tools	TensorFlow, SQL, CST Studio Suite, AutoCAD, Mathematica, C++, NI LabVIEW, 3D printing, Power BI
Core competencies	Materials characterization, nanoscale metrology, scientific instrumentation, R&D technical reporting

PROFESSIONAL EXPERIENCE

Research Technician - Institute of Micro and Nanotechnology, IMN-CNM, CSIC | Madrid, Spain | Feb 2022 - Present

- Develop advanced AFM-based characterization methods for nanomaterials, semiconductors, biosensors and biological systems, supporting multidisciplinary research in materials physics, nanoscience and biophysics.
- Operate, calibrate and maintain advanced instrumentation including AFM, SEM-EDX, TEM, FIB lithography, XRD, HPLC-MS, Raman and FTIR, ensuring reliable availability for multiple research teams.
- Translate experimental requirements into reproducible protocols, datasets and technical reports, improving traceability and usability of results across R&D projects.
- Co-author 10+ peer-reviewed publications in materials characterization, nanoscience, biosensors and applied research.
- Contribute to national, international and industry-linked R&D projects, including qNanoFoPho, a 3-year AFM-confocal nanoscale metrology project with a €200K direct-cost budget, plus biosensor/autism-related initiatives.
- Characterize thermoelectric materials through electrical and thermal transport measurements using Hall, Seebeck, NanoTR and PicoTR systems.
- Support ISO 9001 quality processes through equipment procedures, audit preparation, workflow traceability, SOP improvement and GLP/GMP-aligned laboratory documentation.
- Develop computational tools for scientific data analysis, numerical modelling and experimental reporting, reducing recurring manual processing time and improving reproducibility.
- Manage project-related procurement, supplier coordination and budget tracking to support timely delivery of research milestones.
- Train junior researchers on analytical methods, sample preparation and instrumentation workflows.

Research Assistant, Materials Physics - SMAP Group, Surfaces and Porous Materials | Valladolid, Spain | Jan 2021 - Sep 2021

- Synthesized and characterized 10+ mixed-matrix membrane and polymer formulations for gas-separation applications, optimizing permeability/selectivity/performance trade-offs for CO2 and greenhouse-gas capture use cases.
- Managed laboratory operations, instrument maintenance and safety protocols in a high-volume materials research environment.

EDUCATION

PhD Candidate in Condensed Matter Physics, Nanoscience and Biophysics - Autonomous University of Madrid, Spain | Expected: 2027
Master's Degree in Physics, Materials Physics - University of Valladolid, Valladolid, Spain | 2021
Bachelor's Degree in Physics - University of Valladolid, Valladolid, Spain | 2020

CERTIFICATIONS & TRAINING

- Data Manipulation in Python, Data Science, Python for Machine Learning and Generative AI - Harvard University / Coursera, 2025
- Financial Modeling for Data Scientists; Data Analysis and Big Data / AI Through Visualization - Coursera / Columbia University, 2025
- Data Science and Data Analysis with R / SQL for Data Analysis and SQL Database Design - Coursera, 2025 / Udemy 2024
- Computational Physics: Mathematica, C++ and NI LabVIEW - University of Valladolid, 2021

LANGUAGES

Spanish: Native | English: B2